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NOUS Platform Architecture

Strategic Framework for Integrated Cognitive-Linguistic Health Surveillance with Real-Time Deployment

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SECTION 01

Platform Philosophy

The NOUS Platform is the world's first comprehensive cognitive-linguistic health surveillance infrastructure — a unified system that assesses, monitors, and forecasts human health through the medium of language. Built on Neural Oscillation Signature Theory (NOST), the platform operates across four architectural layers: a foundational technology core, a diagnostic layer spanning five clinical domains, an intelligence layer comprising three cross-domain analytical modules, and a real-time deployment layer (RTCA) that extends the entire processing pipeline into live streaming audio. Each layer compounds the value of the others, creating a data flywheel that no standalone product can replicate.

> **The strategic principle is simple:** NOUS is the platform. The modules are the products. The platform is the moat. Every analysis across every module feeds the same analytical engine. Every customer activation deepens the data substrate. Every new domain or module extends the value of every existing one. This is precision medicine at the phenotypic level — assessing what the brain is actually doing, through the highest-bandwidth output humans produce.

SECTION 02

System Architecture

NOUS PLATFORM — LAYERED ARCHITECTURE

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INTELLIGENCE LAYER

Cross-Domain Analytical Modules

THANATOS

Mortality Forecasting

Predictive life-course surveillance through multi-domain NOSS decay trajectory analysis. Forecasts mortality risk at 6/12/36-month horizons.

Δ-NOSS MCI TLMP

ORACLE

Pharmacological Response

Precision medicine through pre-treatment cognitive-linguistic biomarker analysis. Predicts drug efficacy, adverse reactions, and optimal dosing.

PRI ARS TWE TTM

AEGIS

Cognitive Fitness Certification

On-demand, non-invasive cognitive performance verification for safety-critical roles. 3–5 minute assessment, real-time certification.

CPS FFD IDA LPB

↑ NOSS Data Flows Up · Intelligence Flows Down ↓

DIAGNOSTIC LAYER

Clinical Domains (NOSS Classification)

DOMAIN 1

Mental Health

Depression, anxiety, psychosis, PTSD, bipolar, personality disorders, substance use

DOMAIN 2

Cerebrovascula

Stroke risk, TBI, neurodegeneration , dementia, cognitive decline

DOMAIN 3

Cardiovascular

Cardiac risk, hypertension markers, arrhythmia indicators, vascular health

DOMAIN 4

Oncology

Cancer-related cognitive signatures, treatment response, recurrence monitoring

DOMAIN 5 · GENESIS

Prenatal & Maternal-Fetal

Maternal health surveillance, fetal development markers, prenatal risk assessment, stillbirth and maternal mortality forecasting

↑ Raw Linguistic Data Flows Up · Calibration Flows Down ↓

DEPLOYMENT LAYER

Input Pathways & Streaming Modes

AAE

Adaptive Assessment Engine

Structured assessment orchestrator. Accepts

RTCA

Real-Time Call Analysis

Streaming deployment adapter. Converts live audio from telehealth, call

VOCA

Continuous Vocal Monitoring

Persistent vocal baseline tracking with adaptive deviation detection.

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narrative input, routes to REMIEL, detects data gaps, generates adaptive follow-ups. Controls assessment cadence and diagnostic provocation.

Structured Input
Diagnostic Provocation
Sector Isolation

centers, crisis lines, and operational settings into continuous NOUS pipeline input with sub-2-second end-to-end latency.

Streaming Audio
Speaker Diarization
Live NOSS
Edge-Cloud

Monitors cognitive state between sessions through learned individual voice profiles.

Vocal Baselines
Deviation Detection
Alerts

Feature Vectors Flow Up · Patient Context Flows Down

FOUNDATION LAYER

Core Technology Infrastructure

NOST
Neural Oscillation Signature Theory

REMIEL
AI Processing Engine

R2X
Data Pipeline & Analytics

EMPYREAN 3D
Visualization System

NOSS
Classification System

DREAMWEAVER
Dream Analysis Engine

MEMORIA SONATA
Audio Output System

BEVERLY INDEX
Scoring System

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Intelligence Module Profiles

Each intelligence module consumes NOSS outputs from the diagnostic layer and generates higher-order predictive analytics. Modules are commercially independent — customers activate only what they need — but architecturally integrated, sharing data substrates that improve accuracy across the entire platform.

THANATOS

Mortality Forecasting

CORE CONSTRUCTS

- Δ-NOSS temporal decay analysis
- Mortality Convergence Index (MCI)
- Terminal Linguistic Marker Profile
- 6/12/36-month risk stratification

TARGET MARKETS

- Life insurance & actuarial
- Palliative & hospice care
- Pharmaceutical clinical trials
- Elder care monitoring

Year 3: $15–25M | Year 5: $68–100M

ORACLE

Pharmacological Response

CORE CONSTRUCTS

- Pharmacological Receptivity Index
- Adverse Response Signature
- Therapeutic Window Estimation
- Treatment Trajectory Modeling

TARGET MARKETS

- Pharma trial patient stratification
- Clinical decision support (SaaS)
- Health system enterprise licensing
- Psychiatry & neurology practices

Year 3: $12–20M | Year 5: $47–70M

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AEGIS

Cognitive Fitness Certification

CORE CONSTRUCTS

- Cognitive Performance Score (CPS)
- Fitness-for-Duty Thresholds
- Impairment Detection Array
- Longitudinal Performance Baseline

TARGET MARKETS

- Defense & military (DoD)
- Commercial aviation (FAA)
- Healthcare systems (surgical)
- Commercial fleet & transportation

Year 3: $18–30M | Year 5: $81–130M

SECTION 04

Real-Time Call Analysis Module

The RTCA Module is a cross-domain deployment capability that extends the entire NOUS processing pipeline into live streaming audio. It does not introduce new science — it adapts the existing REMIEL extraction, NOST oscillatory mapping, and NOSS classification system for continuous operation on real-time conversational input. RTCA transforms every telephone call, telehealth session, and operational interaction into a five-domain cognitive surveillance session.

STREAMING ARCHITECTURE

Technical Core

PIPELINE COMPONENTS

- Audio ingestion (SIP/VoIP, WebRTC, PSTN, Mobile SDK)
- Pre-processing (noise suppression, speaker diarization, VAD)
- Streaming REMIEL (all 8 channels adapted for windowed processing)
- Live NOST/NOSS (rolling 12-sector scoring every 15 seconds)
- Real-time alert & output layer (in-call overlay, push notifications, post-call summary)

KEY SPECIFICATIONS

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- 30-second rolling analysis window, 15-second overlap
- 5-second prosodic micro-window for cardiac Pathway 3
- End-to-end latency: <2 seconds (audio → alert)
- Edge-cloud hybrid: feature-only transmission (~2KB/window)

DEPLOYMENT SCENARIOS

Market Applications

PRIMARY DEPLOYMENTS

- Telehealth platform integration (WebRTC SDK)
- Health plan call centers (SIP/VoIP tap)
- 988 crisis line real-time risk stratification
- Oncology post-cycle assessment calls

EXTENDED DEPLOYMENTS

- Sports post-game media analysis (TBI/PTSD screening)
- Military mission debrief cognitive assessment
- AEGIS live fitness certification via routine calls
- GENESIS prenatal surveillance during OB check-ins

AAE INTEGRATION

Sentinel + Investigator

BIDIRECTIONAL ARCHITECTURE

- **RTCA → AAE (Upward):** Ambient NOSS signals detected during live calls trigger AAE to schedule targeted structured assessments
- **AAE → RTCA (Downward):** AAE controls RTCA sensitivity thresholds, sector priorities, and alert rules per patient
- **Conversational Steering:** AAE suggests clinician prompts during RTCA sessions to elicit richer diagnostic signal
- **Lifecycle Manager:** AAE evolves from session orchestrator to continuous cognitive surveillance manager

Year 3: $8.5–14M | Year 5: $48–72M

SECTION 05

Cross-Module Synergies

The integrated architecture creates compounding value: each module's outputs enrich the others, generating data flywheels that standalone products cannot replicate. These synergies deepen

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competitive moats and increase switching costs for every customer engagement.

ORACLE→THANATOS

Treatment response data from ORACLE informs THANATOS mortality trajectories — can pharmacological intervention alter a patient's predicted decline curve?

THANATOS→ORACLE

Mortality risk scores prioritize which patients most urgently need optimized pharmacological selection, creating clinical triage intelligence.

AEGIS→ORACLE

AEGIS detects real-time medication side-effect signatures, feeding back to ORACLE's Adverse Response Signature library with unprecedented scale.

ORACLE→AEGIS

ORACLE flags medications likely to degrade CPS scores, enabling AEGIS to adjust fitness thresholds for medicated personnel proactively.

AEGIS→THANATOS

Longitudinal Performance Baselines from healthy populations create normative datasets that sharpen THANATOS's decay curve detection in clinical populations.

GENESIS→ORACLE

Prenatal Domain 5 data extends ORACLE to maternal pharmacological safety — predicting fetal response to maternal medications through the mother's linguistic biomarkers.

RTCA→AAE

RTCA operates as an ambient sentinel — detecting emerging NOSS signals during live calls that trigger AAE to schedule targeted structured assessments. Sentinel detects; investigator diagnoses.

AAE→RTCA

AAE controls RTCA surveillance parameters per patient — adjusting sensitivity thresholds, sector monitoring priorities, and alert escalation rules based on the patient's established trajectory.

RTCA→AEGIS

RTCA enables continuous AEGIS live certification through routine communication channels — every operational check-in becomes a fitness-for-duty assessment without cooperative testing.

RTCA→ORACLE

Post-medication follow-up calls analyzed via RTCA produce real-time pharmacological response data at population scale, exponentially expanding ORACLE's Adverse Response Signature library.

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SECTION 06

Consolidated Revenue Projections

Intelligence layer module revenue and RTCA deployment revenue are incremental to existing NOUS Platform diagnostic revenue from Domains 1–5. RTCA introduces per-minute and per-encounter pricing models that unlock volume-based revenue at call center scale. All projections assume regulatory clearance timelines as specified in individual module white papers.

MODULE	YEAR 3	YEAR 4	YEAR 5
THANATOS — Mortality Forecasting	$15–25M	$34–55M	$68–100M
ORACLE — Pharmacological Response	$12–20M	$25–42M	$47–70M
AEGIS — Cognitive Fitness Certification	$18–30M	$42–73M	$81–130M
RTCA — Real-Time Call Analysis	$8.5–14M	$22–38M	$48–72M
Intelligence & Deployment Layer Total	**$53.5–89M**	**$123–208M**	**$244–372M**

The compounding thesis: By Year 5, the intelligence and deployment layers project $244–372M in annual revenue — before accounting for diagnostic layer revenue from Domains 1–5 or training and certification income. RTCA's per-minute and per-encounter pricing models unlock volume-based revenue streams that scale with telehealth adoption and call center deployment. The integrated platform architecture ensures that each new customer activation, regardless of entry point, feeds data that improves accuracy and value across all modules simultaneously.

SECTION 07

Intellectual Property Architecture

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Existing Pending Patents

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Foundation layer IP covering NOST, REMIEL, dream pattern recognition, multi-modal analysis, and Beverly Index scoring methodologies.

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Proposed Module Filings

THANATOS (4 patents), ORACLE (5 patents), AEGIS (5 patents) — covering core constructs, system methods, and defensive claims.

5–7

Proposed RTCA Filings

Streaming oscillatory extraction, windowed NOSS scoring, prosodic-autonomic real-time cardiac surveillance, acute cerebrovascular detection from live audio, edge-cloud hybrid architecture, speaker-aware multi-target assessment.

40–42

Total Portfolio (Projected)

Comprehensive coverage from foundational theory through intelligence layer applications and real-time deployment architecture. Trade secrets protect operational parameters, weighting algorithms, and proprietary databases.

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